

$\boxed{0} \rightarrow \underline{\text{ans}}$
 $K = \textcircled{7}$
 $f_0 = -1$
 $l_0 = -1$
 $-1 - (-1) + 1$
 $= \textcircled{1}$

BS
 K
 $K=1$
 count
 $(10 - f_0 + 1)$
 $(f_0 \dots l_0)$
 $1's$
 $[0, 1, 1, 1, 2, 2, 3, 3] \rightarrow \underline{\text{sorted}}$
 $1 = 1$
 $3 = 3 - 1 + 1 = \textcircled{3}$

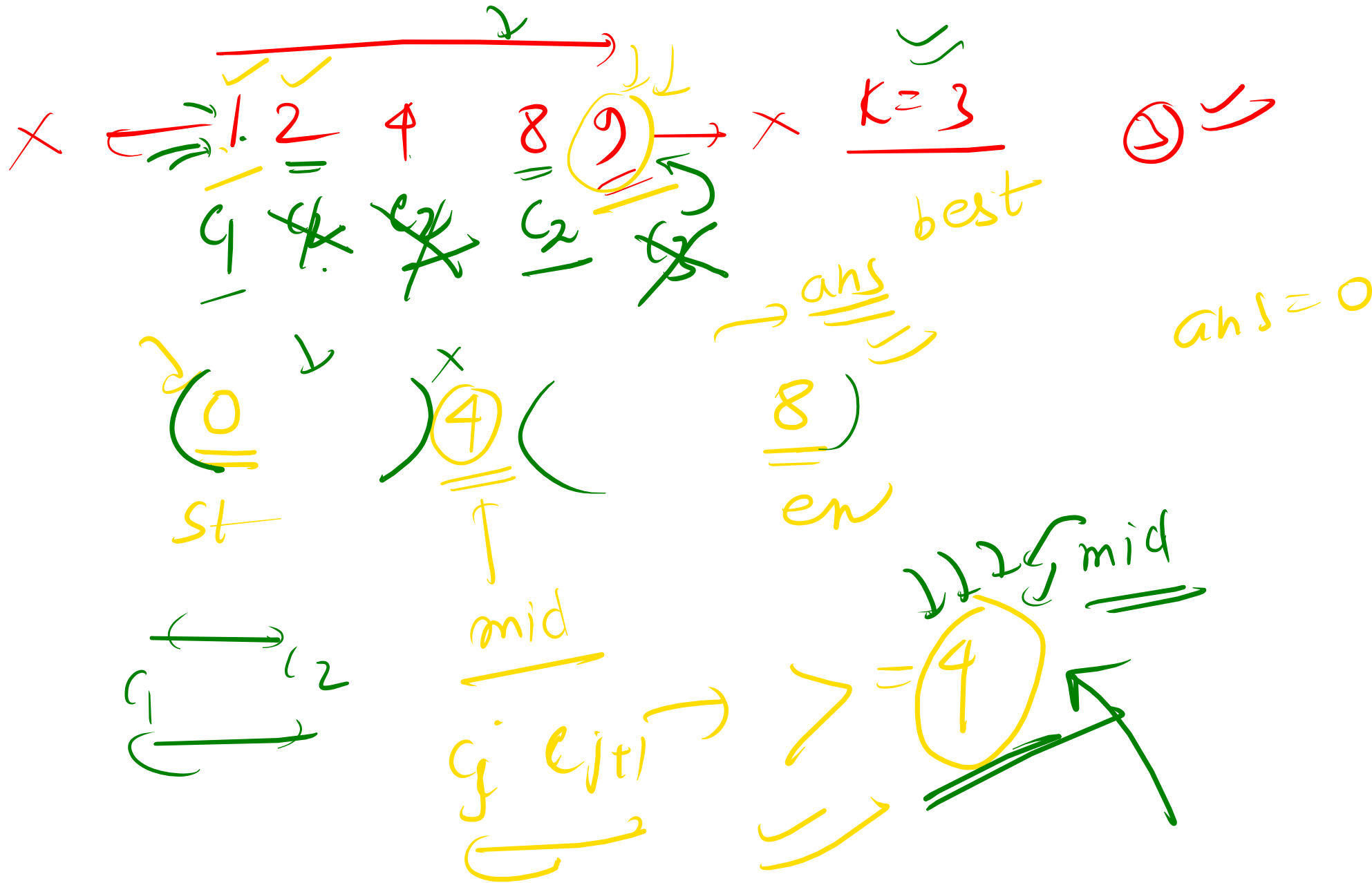
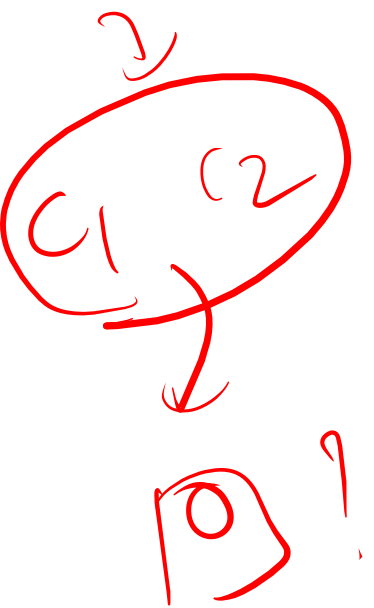
$K=2$

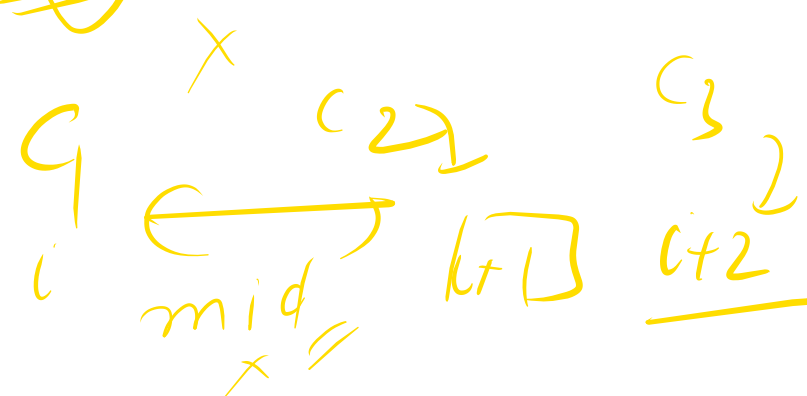
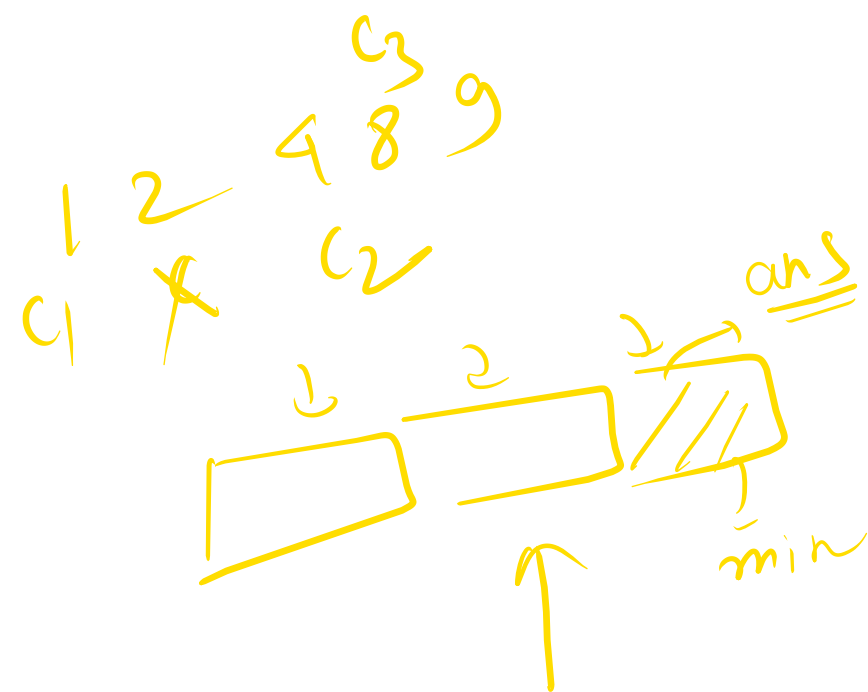
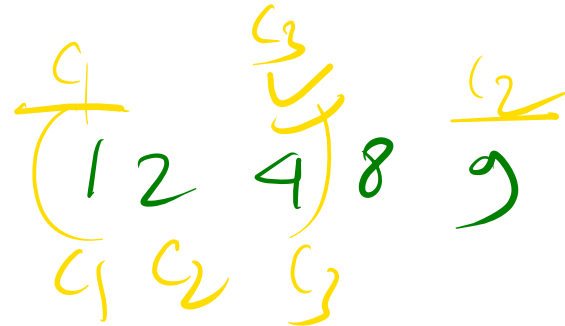
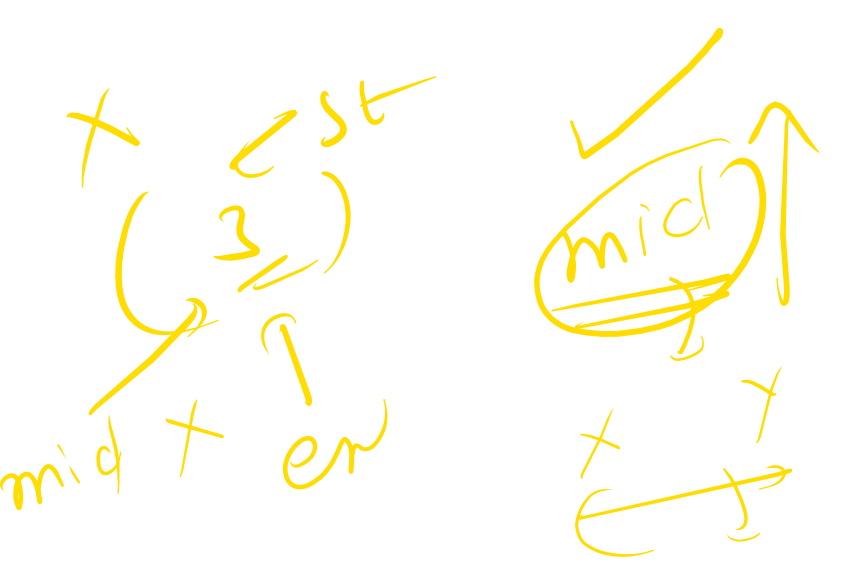
$2 \rightarrow 4$
 $2 \rightarrow 5$

$5 - 4 + 1$
 $= \textcircled{2}$

$$\frac{\underline{f_0} \rightarrow \log(n)}{\underline{f_0} \rightarrow \log(n)} = \cancel{O(\log(n))}$$

$$= \underline{\underline{O(\log(n))}} \checkmark$$





Min $\xrightarrow{\text{suitable}}$ Max
 $\downarrow$
binary ans

Factory Machines (cSES)

ans = ~~18~~ 8
Max
35
en

3 7 $\rightarrow$ min. time
3 2 5
 in which
 7 products
 can be
 produced.

$\downarrow$
 $\left(\begin{smallmatrix} 1 \\ \text{st} \end{smallmatrix} \right)$ mid (18)

5 6 (8)

$2+4+1 = \textcircled{7}$
 $\rightarrow \textcircled{8}$
 $8/3 + 8/2 + 8/5$
6 < 7

8

Java
 $\downarrow$
TLE

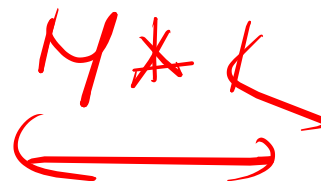
35



—



max time $\rightarrow$ single
product



ans = ~~11~~ 8

$$\begin{array}{r} 24735 \\ 1234 \end{array}$$

 mid ↑
 arr[i]

array div.

$$\begin{array}{r} 24735 \\ 11223 \end{array} \otimes$$

subarray = 1 ... n

K
 ↓
 subarrays
 ↓
 produce

(1)) 11

21

$$\begin{array}{c|c|c} 24 & 73 & 5 \end{array} \rightarrow \underline{K}$$

$$< 11 \quad < 11 \quad < 11$$

$< K \rightarrow (K)$



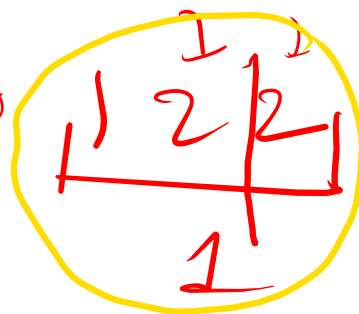
yes
no

$K=9$



$mid = 10 \rightarrow ans?$

10

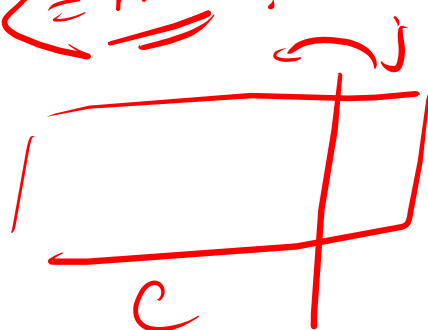


$10 \rightarrow 12 \ 2 \ 10 \ 5$

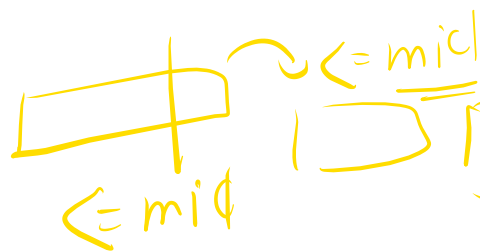
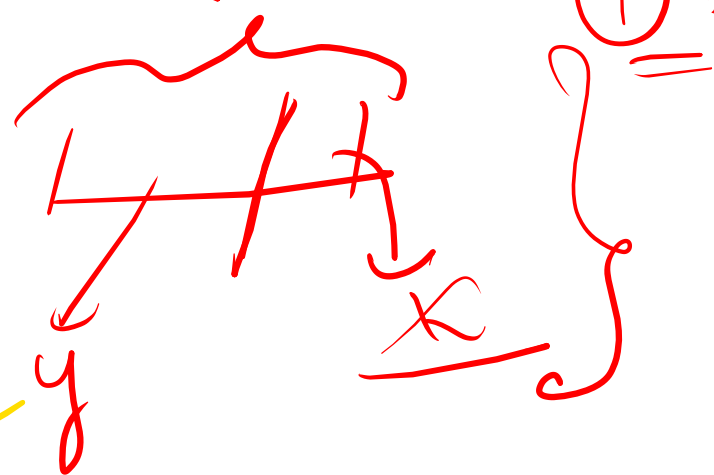
$< K \text{ div.}$

$< = mid$

$s + arr[i] < = mid?$



$< = mid$



(K)

$mid + ans$



segments

$\leq k$ $\otimes$

intuitively

$\frac{2}{\text{mid}}$

$< k$



$i < K^2$

$< k$ $\rightarrow$ (K)

$> k$ $\otimes$



$\leq \text{mid}$

$> \text{mid}$ $\otimes$

en = mid-1

mid $\rightarrow$ ans

mid $\rightarrow$ $\otimes$

st = mid+1